- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

7 The velocity $v\,\text{m s}^{-1}$ of a particle at time t s is given by

$v = 0.5t(7-t)$.

Determine whether the **speed** of the particle is increasing or decreasing when $t = 8$. **[4]**